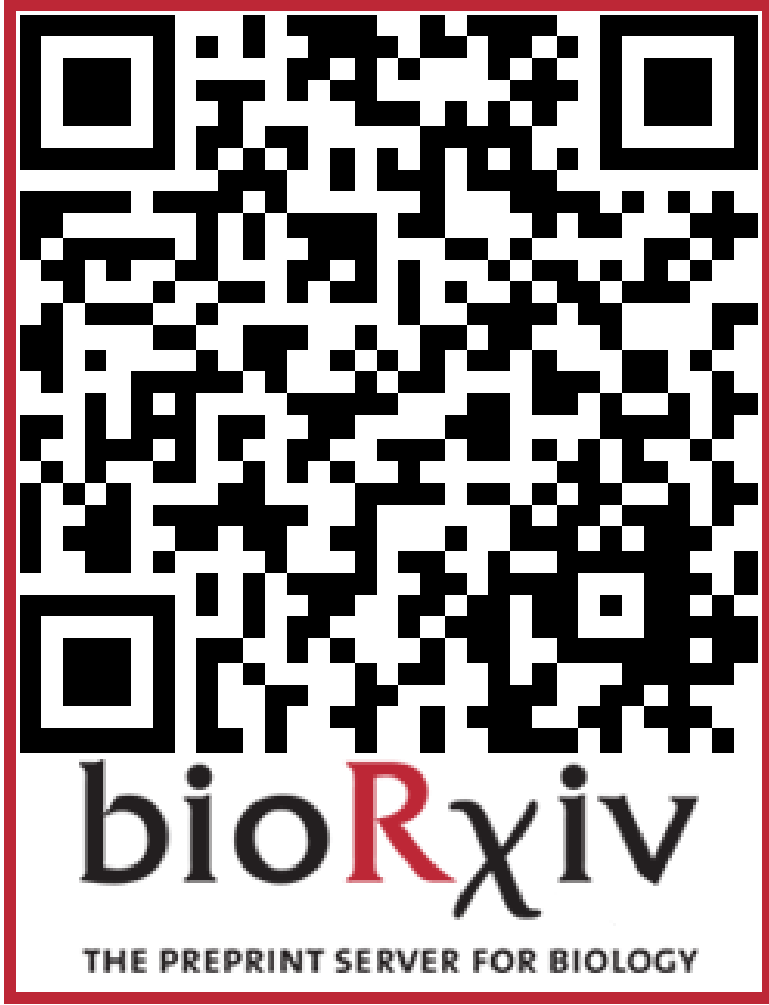


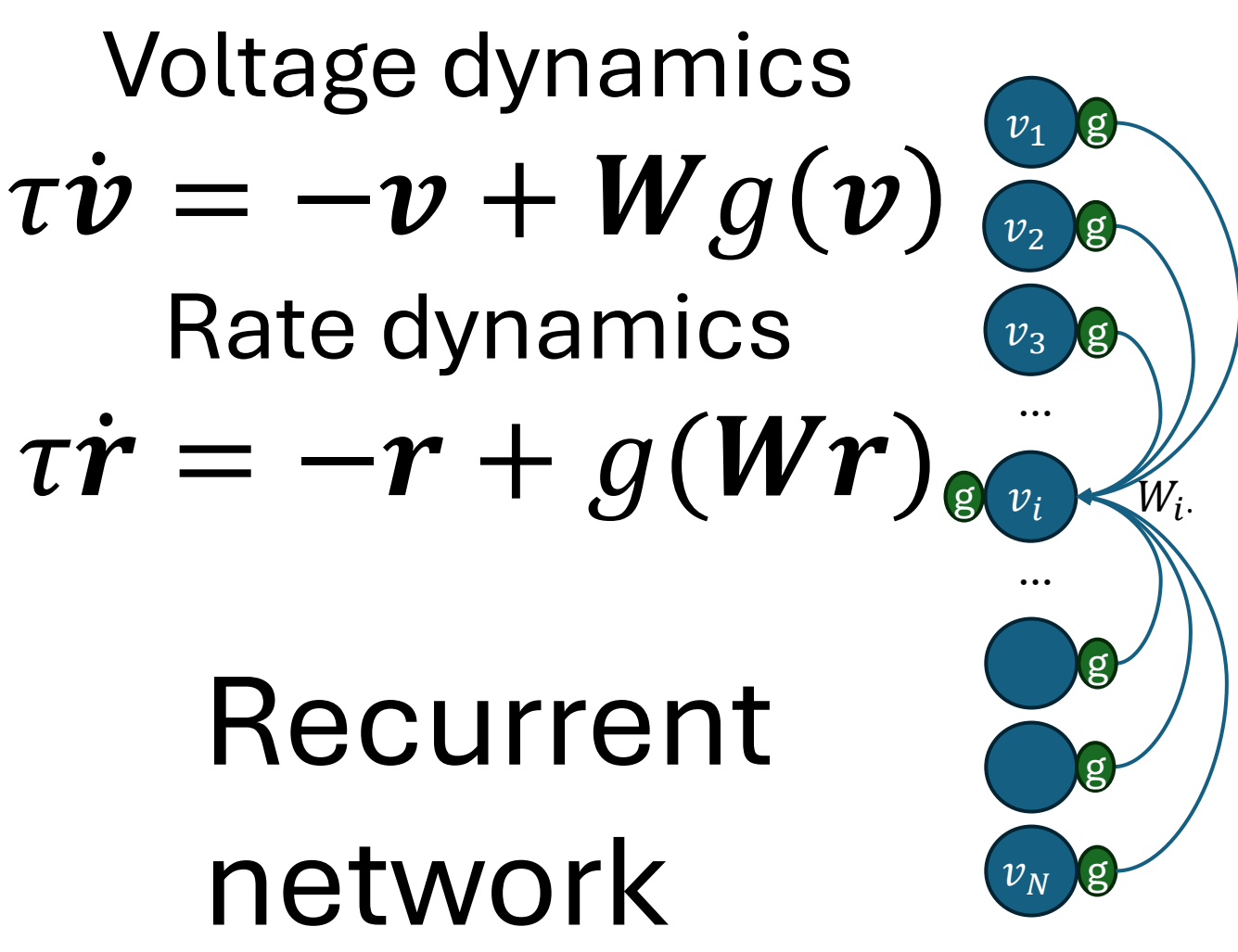
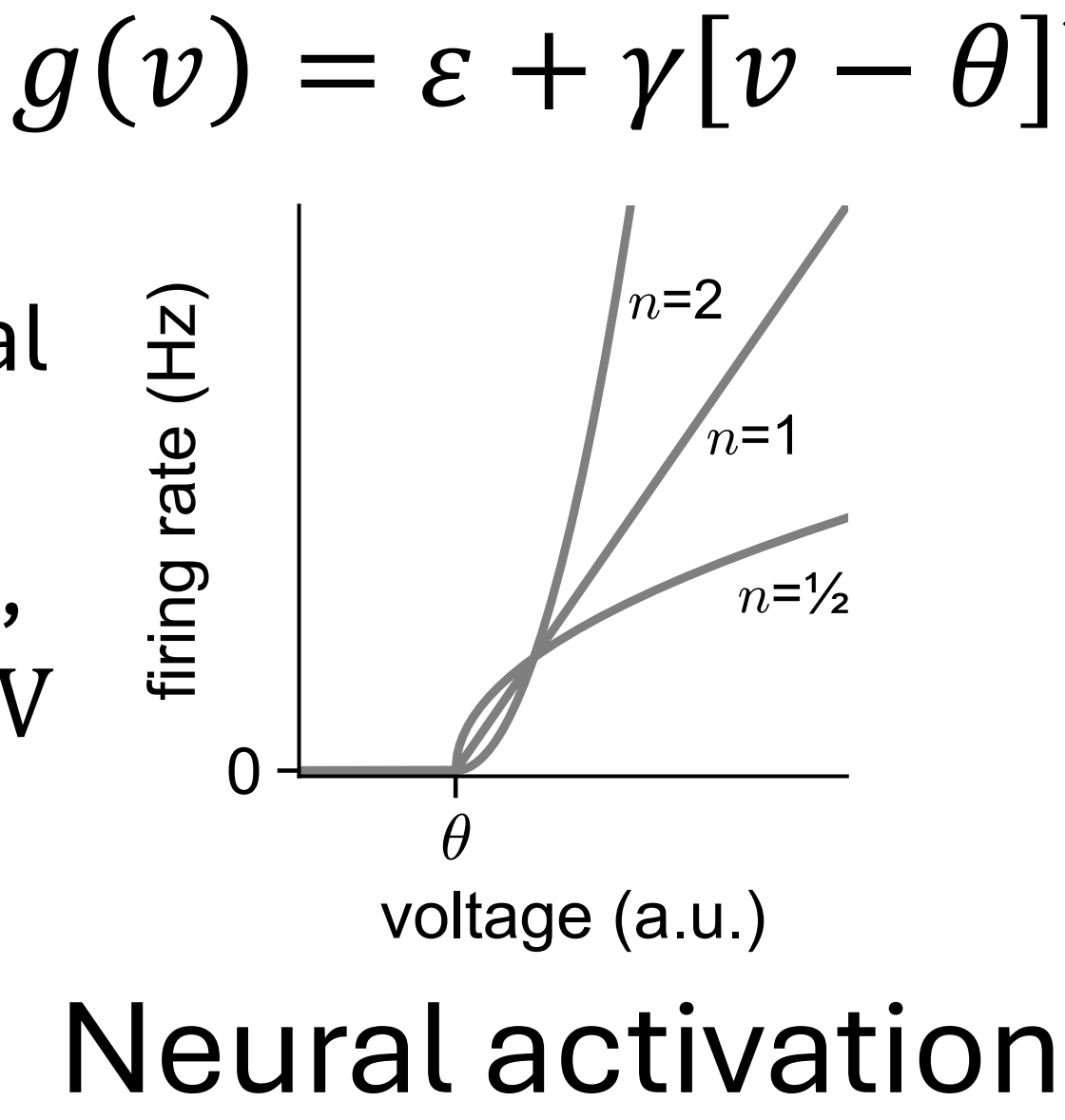
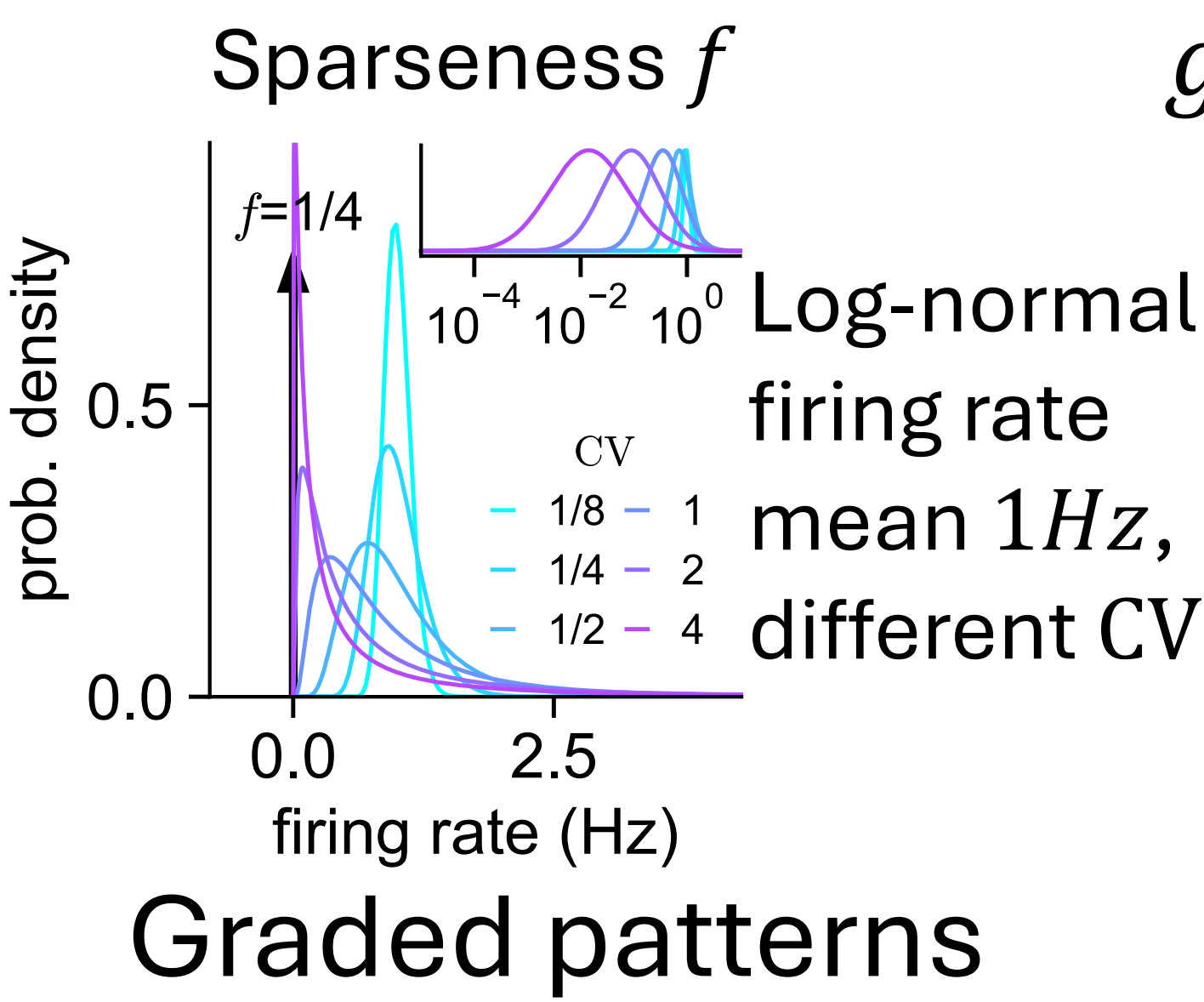
Theory of dynamical stability in attractor neural networks predicts CA3 neurons' activation functions

question

Under what conditions can you store rate-coded memory patterns as stable fixed points of a recurrent neural network?



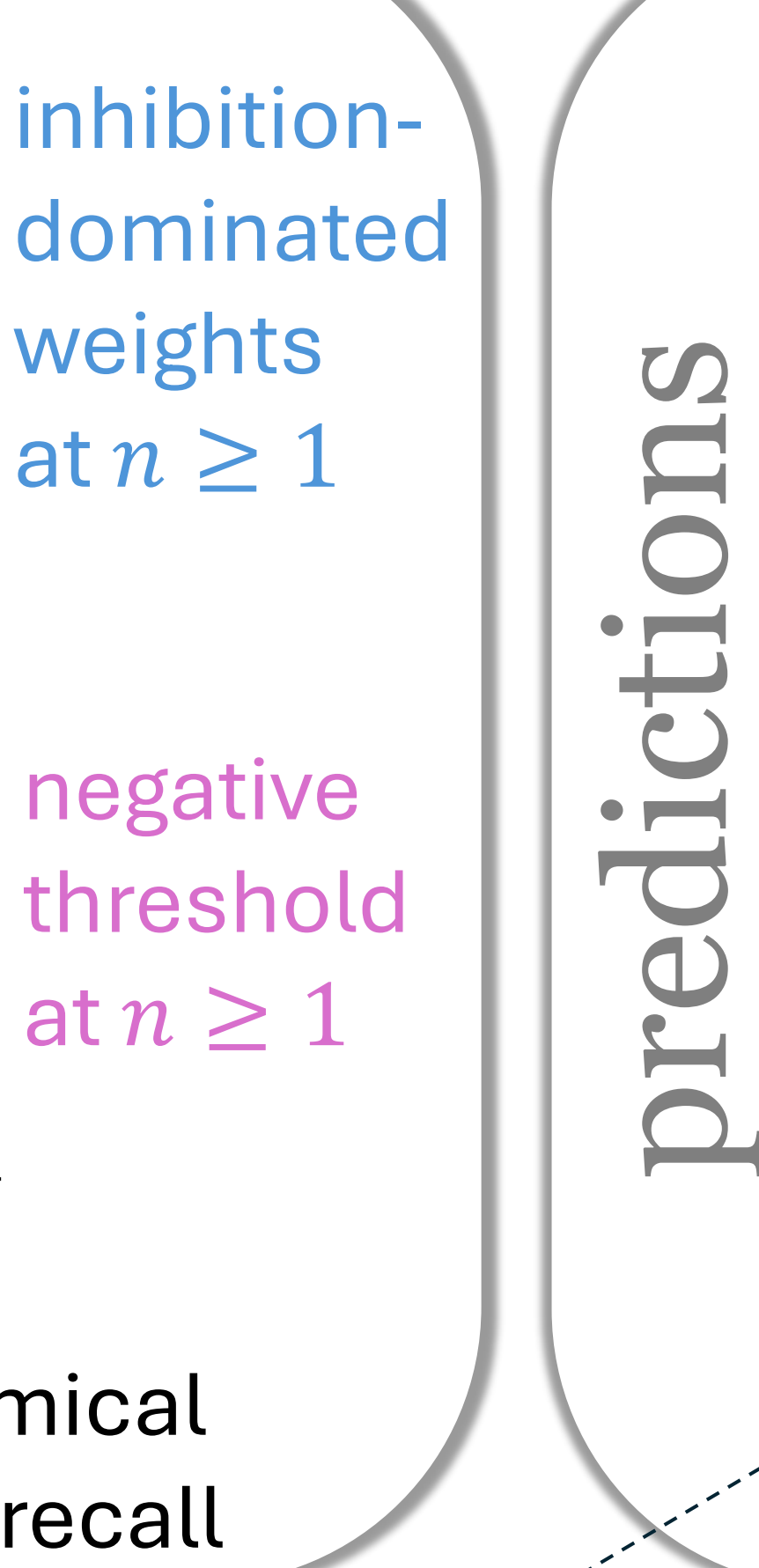
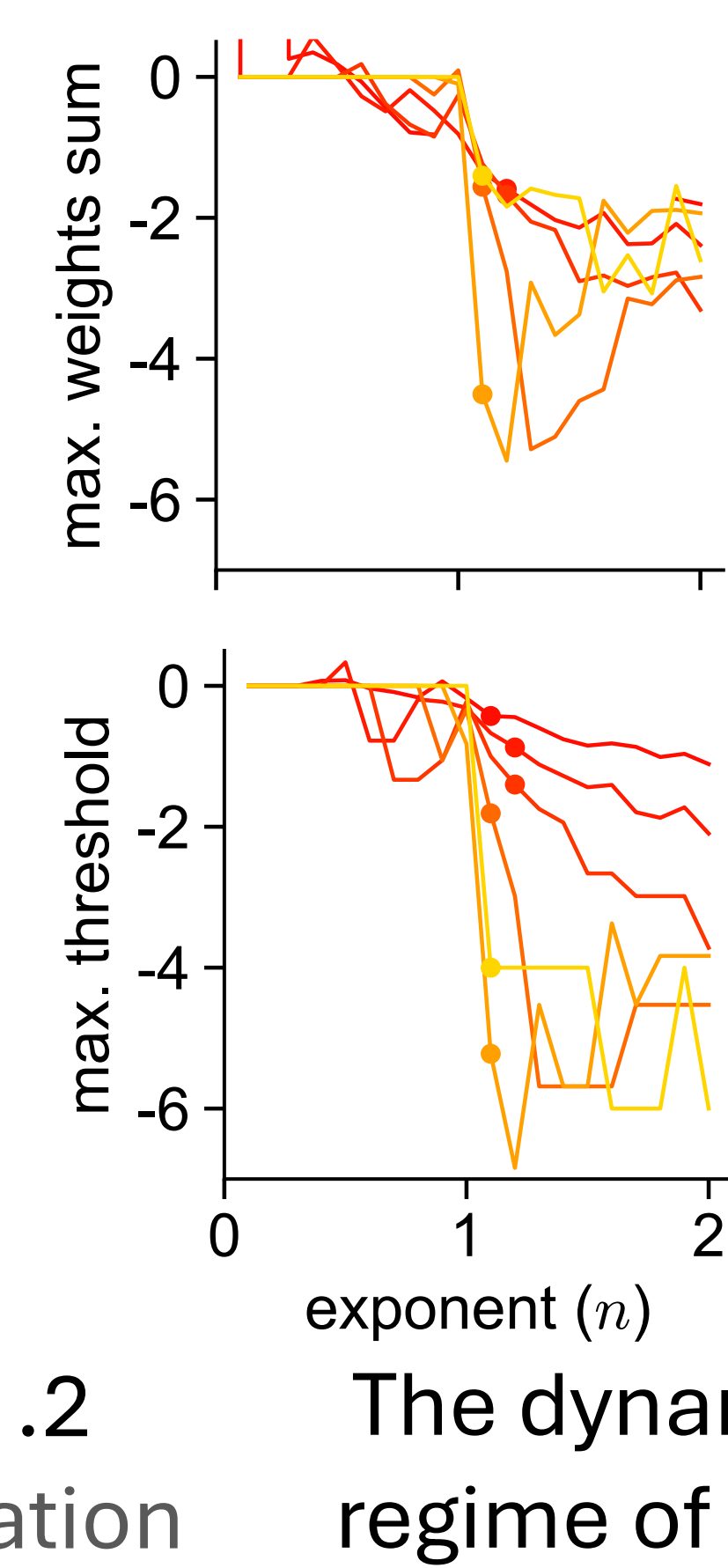
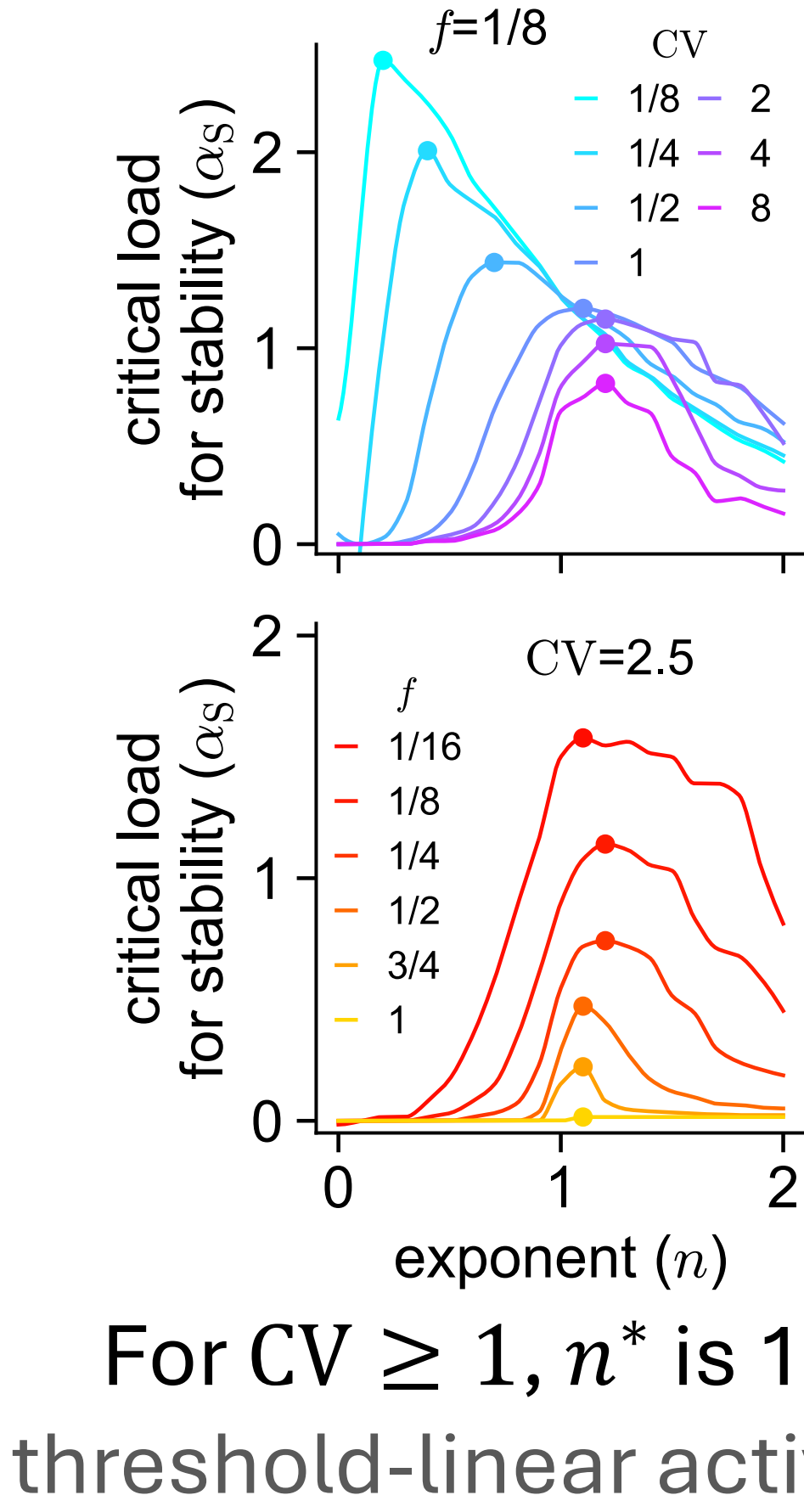
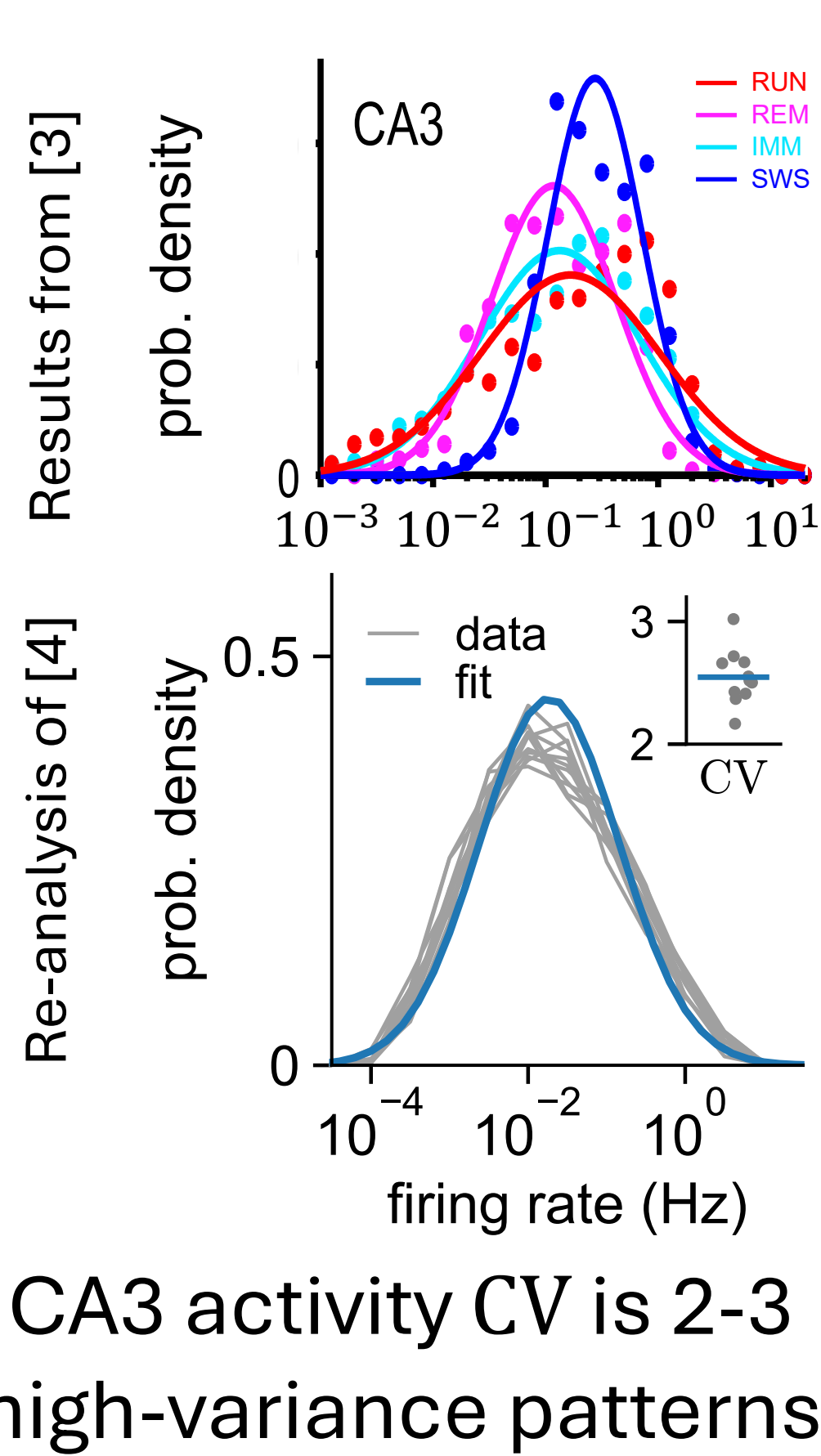
model



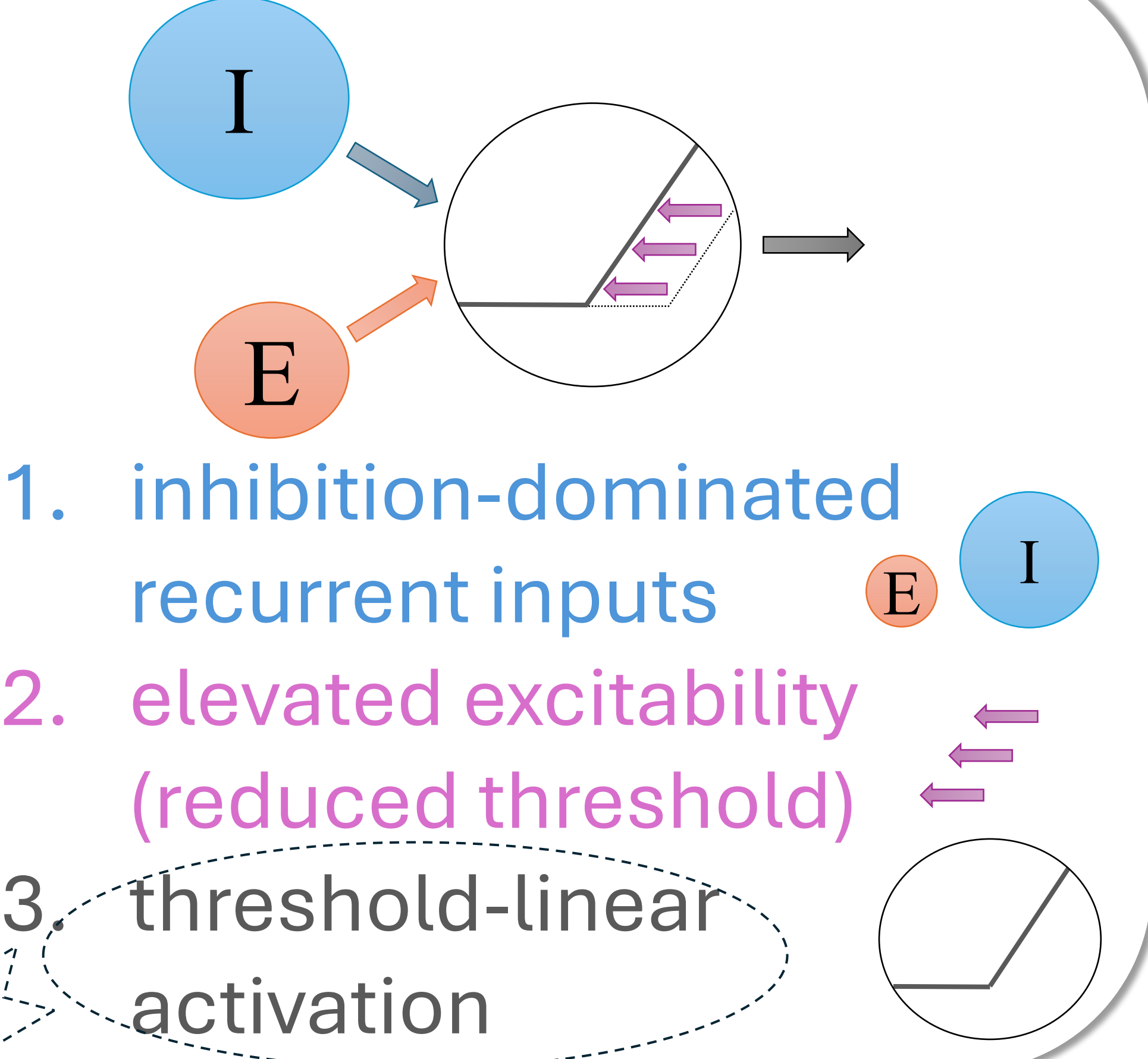
$W^* = \arg \min_W |W|$
 $s. t. r^\mu = g(Wr^\mu) \quad \text{Re} \lambda[J_\mu] < 0$
 $\forall \mu = 1 \dots P$

Storing patterns as fixed points Analyzing stability

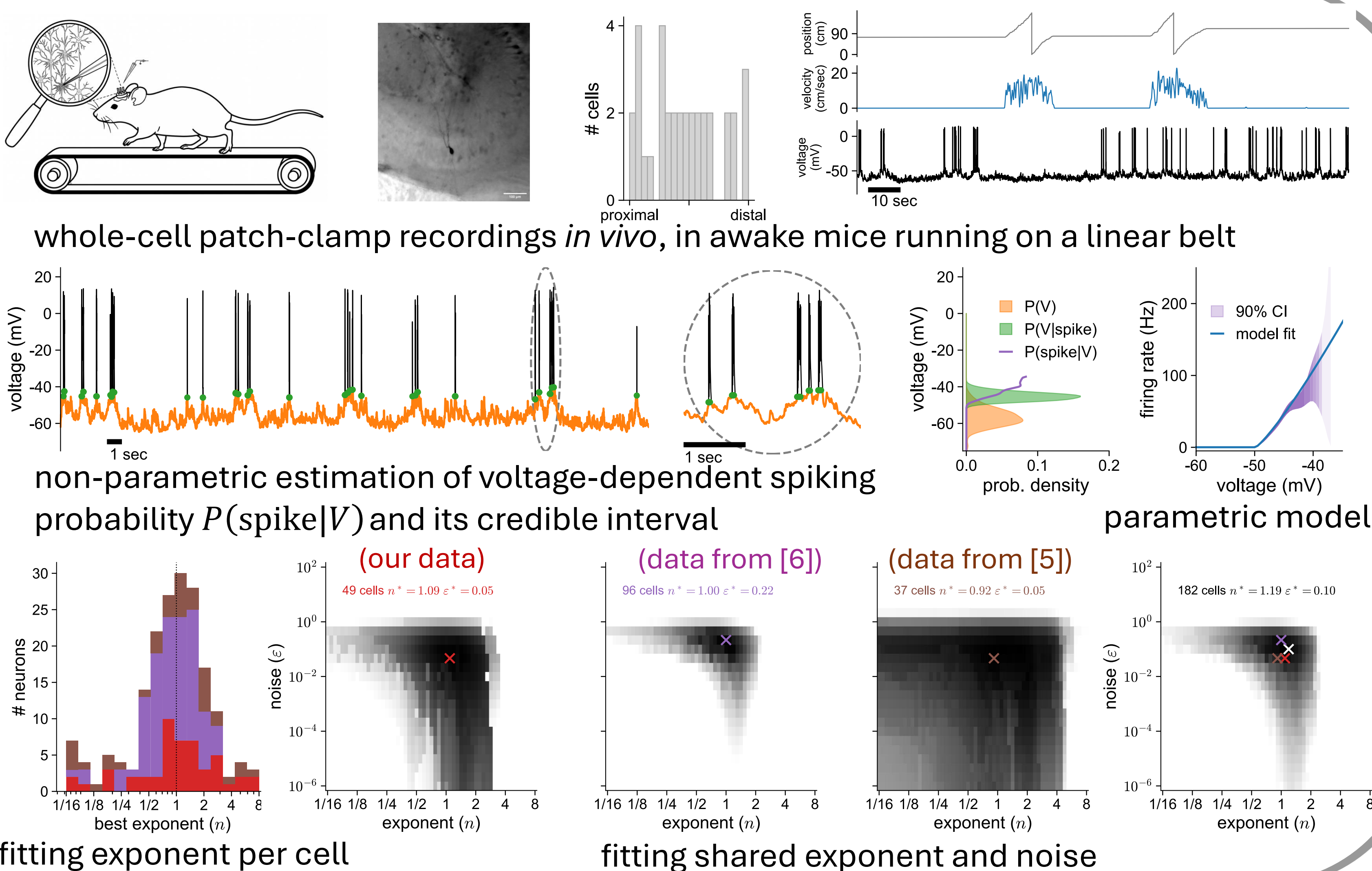
theory



predictions



experimental results



predictions status

1. A net inhibitory response was observed in CA3 pyramidal neurons when activating recurrent collaterals [7], [8], per our “inhibition-dominated recurrent connectivity” prediction.
2. CA3 pyramidal neurons fire spontaneously, without recurrent inputs [9], [10], per our “positive excitability” prediction.
3. Analyzing new data from the Jonas lab and two previous studies [5], [6], we find that an exponent of ~ 1.2 best fits the data, per our “threshold-linear activation” prediction.

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